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How does GPT-2 compute greater-than?: Interpreting mathematical abilities in a pre-trained language model

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Abstract

Pre-trained language models can be surprisingly adept at tasks they were not explicitly trained on, but how they implement these capabilities is poorly understood. In this paper, we investigate the basic mathematical abilities often acquired by pre-trained language models. Concretely, we use mechanistic interpretability techniques to explain the (limited) mathematical abilities of GPT-2 small. As a case study, we examine its ability to take in sentences such as "The war lasted from the year 1732 to the year 17", and predict valid two-digit end years (years > 32). We first identify a circuit, a small subset of GPT-2 small's computational graph that computes this task's output. Then, we explain the role of each circuit component, showing that GPT-2 small's final multi-layer perceptrons boost the probability of end years greater than the start year. Finally, we find related tasks that activate our circuit. Our results suggest that GPT-2 small computes greater-than using a complex mechanism that activates across diverse contexts.

1 Introduction

As pre-trained language models (LMs) have grown in both size and effectiveness, their abilities have expanded to include a wide range of tasks, even without fine-tuning [6]. Such abilities can range from translation to text classification and multi-step reasoning [52]. Yet despite heavy study of these models [41, 42, 26], how LMs implement these abilities is still poorly understood.

In this paper, we study one such LM ability, performing mathematics. Mathematical ability has long been of interest in natural language processing: models have been trained to perform tasks such as simple arithmetic and word problems [51, 46]. Researchers have also fine-tuned pre-trained LMs on these tasks, instead of training from scratch [18, 25]. Recently, however, LMs seem to have acquired significant mathematical abilities without explicit training on such tasks [6, 34, 8, 16].

How these mathematical abilities arise in LMs is largely unknown. While studies have investigated pre-trained LMs' mathematical abilities [32, 38], existing work is behavioral: it explains *what* models can do, rather than *how* they do it. Most work that delves into model internals does so using models trained directly on such tasks: Hupkes et al. [23] probe such models for hierarchical structure, while Liu et al. [27] and Nanda et al. [31] study toy models trained on modular addition. Some studies do examine the structure of number representations in pre-trained models [30, 49]; however, they do not provide a causal explanation about how these models leverage these representations to perform math. The mechanisms underlying pre-trained LMs' mathematical abilities thus remain unclear.

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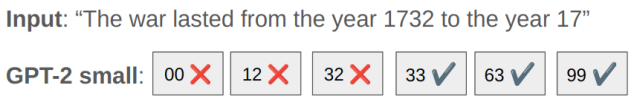

Figure 1: Year-span prediction example (XX=17 and YY=32) with sample (in)valid output years.

To understand the roots of these mathematical abilities, we study them in GPT-2 small[1] [39], which we show still possesses such abilities, despite its small size. This small size enables us to investigate its mathematical abilities at a very low level. Concretely, we adopt a circuits perspective [35, 14], searching for a minimal subset of nodes in GPT-2's computational graph responsible for this ability. To do so, we use fine-grained, causal methods from mechanistic interpretability, that allow us to identify nodes in GPT-2 that belong in our circuit, and then prove our circuit's correctness, through carefully designed causal ablations [21]. We also use mechanistic methods to pinpoint how each circuit component contributes to the mathematical task at hand. The end result of this case study is a detailed description of GPT-2's ability to perform one simple mathematical operation: greater-than.

Our investigation is structured as follows. We first define year-span prediction, a task that elicits mathematical behavior in GPT-2 (Section 2). We give the model input like "The war lasted from the year 1732 to the year 17"; it assigns higher probability to the set of years greater than 32. We next search for the circuit responsible for computing this task, and explain each circuit component's role (Section 3). We find a set of multi-layer perceptrons (MLPs) that computes greater-than, our operation of interest. We then investigate how these MLPs compute greater-than (Section 4). Finally, we find other tasks requiring greater-than to which this circuit generalizes (Section 5).

Via these experiments, we accomplish two main goals. First, we find a circuit for greater-than in GPT-2. This brings new mechanistic insights into math in pre-trained LMs, and builds on the limited existing work on circuits in pre-trained LMs [50] by examining a new task with a wide output space and rich structure. Second, we show that GPT-2's greater-than relies on a complex circuit that activates across contexts. This mechanism surpasses simple memorization, but does not reflect full mathematical competence; it lies between memorization and generalization. We thus add nuance to the memorization-generalization dichotomy, and take the first step towards a rich characterization of the states in between them.

2 Year-Span Prediction in GPT-2

GPT-2's size is ideal for low-level study, especially with potentially resource-intensive techniques like those in Section 3.1. However, this small size poses a challenge: GPT-2 is less capable than larger LMs, which still often struggle with mathematical tasks [29]. With this in mind, we craft a simple task to elicit a mathematical behavior in GPT-2, and verify that GPT-2 produces said behavior.

Task and Dataset We focus on a simple mathematical operation, greater-than, framed as it might naturally appear in text: an incomplete sentence following the template "The <noun> lasted from the year XXYY to the year XX" (Figure 1). The model should assign higher probability to years >YY. We automatically generate sentences using this template. We draw the nouns from a pool of 120 nouns that could have a duration, found using FrameNet [1]; Appendix G lists the full pool of nouns. We sample the century XX of the sentence from $\{11, \ldots, 17\}$, and the start year YY from $\{02, \ldots, 98\}$.

We impose the latter constraints because we want GPT-2 to be able to predict a target as it would naturally be tokenized. However, GPT-2 uses byte-pair encoding, in which frequent strings more often appear as single tokens [44]. Thus, more frequent years—multiples of 100 or those in the 20th century—are tokenized as single tokens; less frequent years are broken into two. This causes a problem: GPT-2 could predict "[00]" after "[17]', but "1700" is always tokenized as "[1700]" in normal data and never as "[17][00]". So, we exclude all single-token years from our year pool. Finally, we want each example to have at least one correct and one incorrect validly tokenized answer, so we exclude each century's highest and lowest validly tokenized year from the pool of start years.

[1]Further references to GPT-2 refer to GPT-2 small

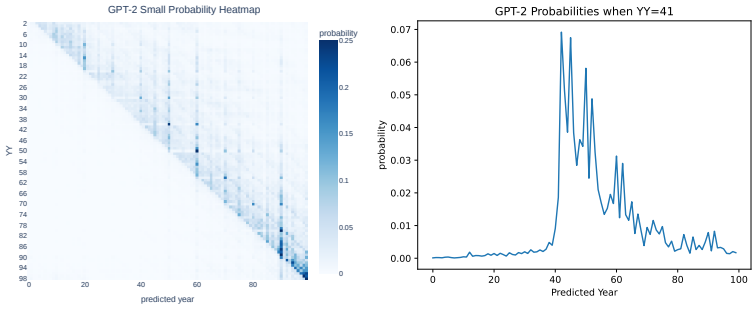

Figure 2: **Left**: Probability heatmap of GPT-2 for year-span prediction. Y-axis: the sentence's start year (YY). X-axis: the two-digit output year candidate. (X,Y): Mean probability assigned by GPT-2 to output year X given input year Y. **Right**: GPT-2's average output distribution when YY=41

Qualitative Evaluation We first qualitatively analyze GPT-2's baseline behavior on this task by running it on a dataset of 10,000 examples. Each example has a noun randomly drawn from our 120 nouns, and a year drawn randomly from the 768 valid years from 1000 to 1899. For each YY in $\{2, \ldots, 98\}$, we take the average of GPT-2's probability distribution over predicted years, for all examples with start year YY; we visualize these average distributions in Figure 2.

GPT-2 appears to perform greater-than on our year-span prediction task: it creates a sharp cutoff between invalid end years ($\leq$ YY), and valid end years ($>$ YY). It assigns higher probability to the latter years, though not all of them: 15-20 years after YY, probabilities drop. The exact length of the year-span receiving higher probability likely reflects patterns in GPT-2's training data. In the real-world, and likely also in GPT-2's training data, our prompts' nouns, such as a "war," "dynasty", or "pilgrimage", have average durations that GPT-2 may have learned, influencing its output.

Quantitative Evaluation We design two numerical measures of model performance for the purpose of quantitative assessment. Let $\texttt{YY} \in \{02, \ldots, 98\}$ be the start year of our sentence, and p_y be the probability of a two-digit output year y. We define the following two metrics:

- **Probability difference**: $\sum_{y > \texttt{YY}} p_y - \sum_{y \leq \texttt{YY}} p_y$
- **Cutoff sharpness**: $p_{\texttt{YY}+1} - p_{\texttt{YY}-1}$

Probability difference verifies that model output reflects a greater-than operation by measuring the extent to which GPT-2 assigns higher probability to years >YY. It ranges from -1 to 1; higher is better. In contrast, cutoff sharpness is not intrinsically connected to greater-than. However, it quantifies an interesting behavior of GPT-2: the sharp cutoff between valid and invalid years. In doing so, it checks that the model depends on YY, and does not produce constant (but valid) output, e.g. by always outputting $p(99) = 1$. Cutoff sharpness ranges from -1 to 1; larger values indicate a sharper cutoff.

We perform this evaluation with our same 10,000-element dataset; on this dataset, GPT-2 achieves 81.7% probability difference (SD: 19.3%) and a cutoff sharpness of 6.0% (SD: 7.2%). Overall, both qualitative and quantitative results indicate that GPT-2 performs the greater-than operation on the year-span prediction task. For more study of GPT-2's behavior on this task, see Appendix A.

3 A Circuit for Year-Span Prediction

Having defined our task, we now aim to understand how our model performs it internally. Since the advent of pre-trained models, many methods, such as attention analysis [9] and probing [3], have sought to answer this question. Probing, which trains auxiliary models (probes) to extract information from model representations, has been particularly popular; it has been used to localize syntactic and semantic processing in many pre-trained LMs [37, 45, 15]. However, it has significant pitfalls: most crucially, probes can sometimes extract information from model representations that is irrelevant to model behavior [13, 40, 22]. To avoid this issue, other work has used causal interventions [36] to intervene on model internals, and observe changes in model behavior [20, 48, 17]. This ensures that our insights about model internals are actually functionally relevant to model behavior.